COMP3311 Week 10 Thursday Lecture

- Week 10 Thursday
- Transactions (recap)
- Concurrency Control in SQL
- Locking in PostgreSQL
- Exercise: Locking
- Locking and Performance
- Multi-version Concurrency Control
- 6 Brief History of Databases
- Future of Database
- Big Data
- Object-relational Mapping
- Information Retrieval
- Multimedia Data
- Uncertainty
- Data Management Systems
- Graph-based Database
- Disjointed Databases
- Non-relational Databases
- Other DB Research
- Course Review: Exam
- COMP3311 Course Aims
- Course Outcomes
- Assessment Summary
- Final Exam
- Exam Tips
- Supplementary Exams
- Assessment
- Course Materials...
- Finale...

◆ Week 10 Thursday

In today's lecture ...

- Concurrency, Database History, Course Review, Exam Preview

Things to do ...

- Assignment 2 due Thursday 24 April 2025 @ 23:59:59 (TODAY!)
- MyExperience course evaluation now open

◆ Transactions (recap)

Transaction are application atomic operations, involving multiple DB ops

A transaction must always terminate, either:

- successfully (**COMMIT**), with all changes preserved
- unsuccessfully (**ABORT**), with database unchanged

pending
reading
active
writing
failed
committed
aborted
COMMIT
ABORT
ABORT
COMMIT
read/write
data lang. number crunch
checkpoint
checkpoint
checkpoint

Work by Jennifer Widom at Stanford on the Trio system

- extends the relational model (ULDB)
- extends the query language (TriQL)

◆ Stream Management Systems

Makes one addition to the relational model

- stream = infinite sequence of tuples, arriving one at a time

Applications: news feeds, telecoms, monitoring web usage, ...

RDBMSs: run a variety of queries on (relatively) fixed data

Stream DBMSs: run fixed queries on changing data (stream)

Approaches:

- window = relation formed from a stream via a rule
- stream data type = build new stream-specific operations

◆ Graph-based Databases

Use graphs rather than tables as basic data structure tool.

Applications: complex data representation, via "flexible" objects

Implementing graphs in RDBMSs is possible, but often inefficient.

Graph nature of data changes query model considerably.

- different kinds of queries: KNN, neighbourhood, skyline

Research problem: query processing for large graphs

◆ Dispersed Databases

Characteristics of dispersed databases:

- very large numbers of small processing nodes
- data is distributed/shared among nodes

Applications: environmental monitoring devices, "intelligent dust" ...

Research issues:

- query/search strategies (how to organise query processing)
- distribution of data (scale-off between centralised and diffused)

Less extreme versions of this already exist:

- grid and cloud computing
- database management for mobile devices

◆ Main-memory Databases

Disk-based data is slow to access, and page-at-a-time

- RDBMS techniques aim to minimise page-read/writes
- methods: buffering, indexes, join algorithms

Non-volatile non-disk storage is becoming larger and cheaper

- many databases would fit in memory ... disk as a backup/archive

Research problems:

- how to structure data in SSD-like devices
- how to process queries in such an environment

◆ Other DB Research

Open problems:

- database integration/federation
  - making a collection of heterogeneous databases work together
  - schema reconciliation is a difficult problem (recall, schema)
- replication
  - maintain consistent copies of one database across multiple hosts
  - how to propagate changes systematically?
- column-based storage
  - row = collection of values for a given attribute
  - column = collection of attributes for tuple

◆ Beyond COMP3311

COMP9312 Data Analytics for Graphs

- manipulating large-scale graph structures

COMP9331 Big Data Management

- handling data that won't fit in a DBMS on one machine

COMP9315 Database Systems Implementation

- comprehensive study of DBMS internals

COMP9319 Web Data Compression and Search

- compression and searching algorithms, and XML

COMP6714 Information Retrieval and Web Search

- finding information in unstructured text

◆ COMP3311 Course Aims

At the end of this course you should be able to:

- develop accurate, non-redundant data models;
- realise data models as relational database schemas;
- formulate queries via the full range of SQL constructs;
- use stored procedures and triggers to extend DBMS capabilities;
- write applications in Python that interact effectively with databases;
- understand the overall architecture of relational DBMSs;
- analyze performance issues in relational database applications;
- understand how queries are evaluated via relational algebra operations;
- understand the concepts behind transactions and concurrency control

◆ Syllabus Overview

1. Data modelling and database design
   - Entity-relationship (ER) design, relational data model
   - Relational theory (algebra, dependencies, normalisation)
2. Database application development
   - SQL for querying, data definition and modification (PostgreSQL version)
   - extending SQL: Queries, Functions, Aggregates, Triggers
   - PostgreSQL, psql, pgAdmin, PL/pgSQL (procedural SQL)
   - PQXX: interfacing to C++ applications
   - Python3, Psycopg2, accessing data programmatically
3. DBMS theory/technology
   - relational algebra, functional dependencies, normalisation
   - performance tuning, indexing, query optimisation
   - DBMS architecture, query processing, transaction processing

Things in (grey) will definitely **not** be examined.

◆ Assessment Summary

Your final mark/grade will be determined as follows:

```
quizzes  = mark for on-line quizzes    (out of 12)
ass1     = mark for assignment 1       (out of 15)
ass2     = mark for assignment 2       (out of 16)
exam     = mark for final exam         (out of 60)
chXtam   = exam >= 25                  (after scaling)
mark     = quizzes + ass1 + ass2 + exam
grade    = SC|CR|DN|PS if mark >= 50 && chXtam
         = FL          if mark < 50 && chXtam
         = UF          if isXtam
```


◆ Final Exam

Mon 12 May, 3-hour exam, morning and afternoon sessions

Morning: arrive by 9:45am, Afternoon: arrive by 1:00pm

Exam environment:

- invigilated, in the CSE labs, on CSE workstations
- closed exam environment, no Webcms3, no email, no web
- provided questions, documentation, course notes
- no BYO laptop

◆ Final Exam (cont)

Questions ...

- 5 practical exercises, using PostgreSQL/Python3/Psycopg2
- 4 theory exercises, typed answers, like tute questions

What you have access to ...

- the exam paper (html), templates for all questions
- checking scripts for programming questions (like **autotest**)
- documentation for PostgreSQL/Python/Psycopg2

What you do not have access to

- the class directory, your home directory
- Webcms3, email, messaging, etc.

◆ Final Exam (cont)

When you login (z0/z5/ztest) into the exam environment ...

- we create some working directories
  - including templates and testing scripts
- we create a local PostgreSQL server for you
- we load the exam database into your PG server
- we start a web browser, pointing at the exam entry page

PostgreSQL server runs on your workstation

Database: will not be huge, will not be overly complex (e.g. ~ 11 tables)

You do **not** have access to VLab or /localstorage or /home/HOU/

◆ Final Exam (cont)

For the prac questions ...

- files are contained in a **q3** directory
- including a template, test cases and a **check** script

For the theory questions ...

- each question has a text file template **q7.txt**

General suggestions ...

- submit your answers using a **submit** script (e.g. **submit q3**)
- give up on questions that are taking too long (e.g. ~ 30 mins for prac)
- submit questions as you complete them; can submit multiple times
- don't leave all submissions to the last minute; if late, not marked

◆ Final Exam (cont)

Want to know what are the questions on the exam ...?

◆ Final Exam (cont)

Questions are as follows:

1. an SQL query
2. another SQL query
3. a PL/pgSQL function
4. a Python3/Psycopg2 script
5. another Python3/Psycopg2 script
6. an analysis/synthesis question
7. another analysis/synthesis question
8. etc. etc. etc.

Note that the database for Q1-5 will be available in advance.

◆ Revision

Sources for revision material:

- Lecture Material: Prac/Tute Exercises, Assignments
- Fundamentals of Database Systems, Elmasri/Navathe
- Database System Concepts, Silberschatz/Korth/Sudarshan
- Database Systems, Garcia-Molina/Ullman/Widom
- Database Systems: Complete Book, Garcia-Molina/Ullman/Widom
- Database Systems: Agro-oriented, Kifer/Bernstein/Lewis
- PostgreSQL/Python3/Psycopg2 Documentation (to some extent)

◆ Supplementary Exams

Supplementary Exams are only available to people who

- are absent from the Final Exam with good reason
  (good = documented, serious, clearly affects ability to do exam)
- have performed well during the rest of the semester

If you are awarded a Supp Exam ...

- to be held in O week of 2T12
- you **must** make yourself available for it
- non-attendance at the Supp → mark of 0 for the exam

◆ Assessment

Assessment is about determining how well you understand the syllabus.

If you can't demonstrate your understanding, you won't pass.

In particular, I don't pass people just because ...

- please, please, ... my parents will be ashamed of me
- please, please, ... I tried really hard on this course
- please, please, ... this is my final course to graduate
- please, please, ... I'll be excluded if I fail COMP3311
- please, please, ... if I fail this, I can't do COMP9 3xx
- etc. etc. etc.

◆ Assessment (cont)

Of course, assessment isn't a "one-way street" ...

- we get to assess you in the final exam
- you get to assess us in the Course Evaluation

MyExperience evaluations are online (via MyUNSW) NOW

Several evaluations: course, lecturer, tutor

Telling us good things is ok.

Telling us things to improve is very useful.

◆ Some Thoughts ...

You need to learn for life, not just the exam.

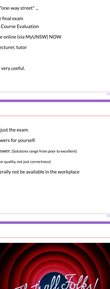
In particular, learn to find answers for yourself.

Typically, no single correct answer. (Solutions range from poor to excellent)

Take pride in your work. (Aim for quality, not just correctness)

VScode and autotest will generally not be available in the workplace

◆ Finally ...



Good Luck with the Exams ... and Life ...